DHCP and NAT Troubleshooting Lab Report

Activity 7.4.3

Author: Aymane Es-salehy
Date: July 16, 2026

Diagnosis and correction of DHCP, routing, and NAT faults in a three-router network.

1 Introduction

The network started in a broken state: no WAN connectivity between R1 and R2, DHCP not handing out addresses, no route to the internet from R1, and NAT misconfigured on R2. The goal was to restore full connectivity — internal clients reaching the ISP, and the ISP reaching the DMZ web server — without rebuilding the configuration from scratch.

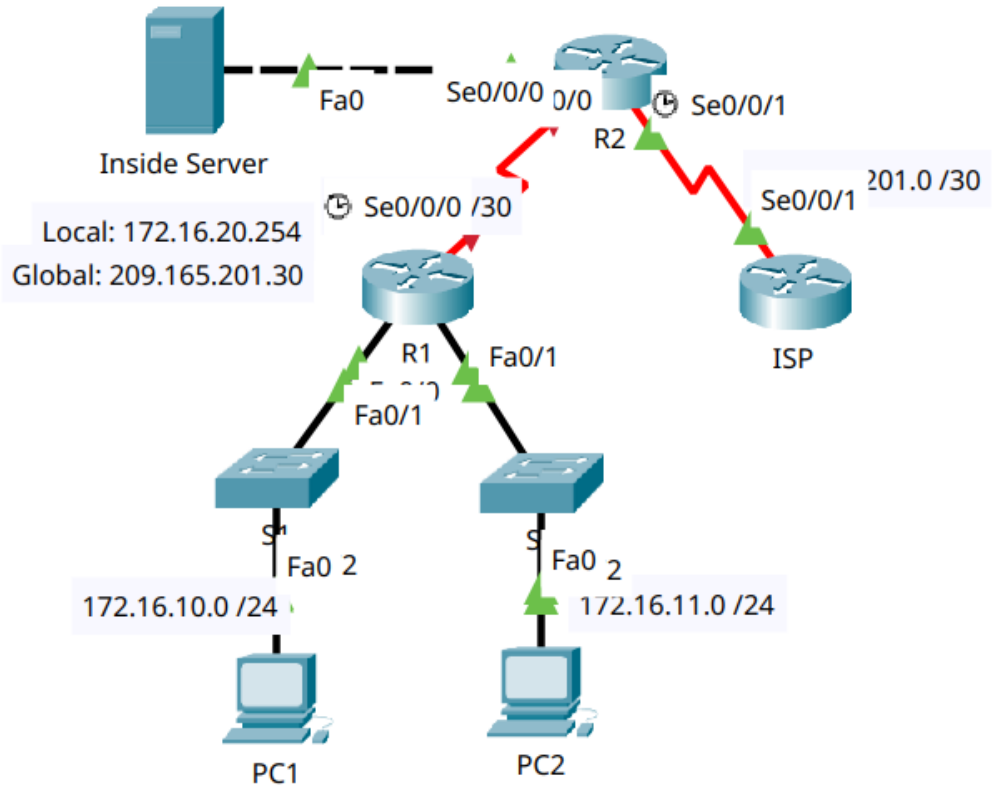


Figure 1: Network topology

Device	Interface	IP Address	Mask
R1	S0/0/0	172.16.0.1	/30
R1	Fa0/0	172.16.10.1	/24
R1	Fa0/1	172.16.11.1	/24
R2	S0/0/0	172.16.0.2	/30
R2	S0/0/1	209.165.201.1	/30
R2	Fa0/0	172.16.20.1	/24
ISP	S0/0/1	209.165.201.2	/30
Web Server	Local	172.16.20.254	/24
Web Server	Public (NAT)	209.165.201.30	–

2 Troubleshooting

2.1 WAN link down

`show ip interface brief` on R1 showed the serial link to R2 was up physically but the line protocol was down — the link was not passing traffic (Figure 2). Pings from R1 to R2 and beyond failed completely (Figure 3).

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        172.16.10.1     YES manual up                    up
FastEthernet0/1        172.16.11.1     YES manual up                    up
Serial0/0/0            172.16.0.1      YES manual up                    down
Serial0/0/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Figure 2: R1 interfaces — serial link not passing traffic

```
R1#ping 209.165.201.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.201.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R1#ping 172.16.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.0.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

Figure 3: R1 pings failing before the fix

Fix: re-enabled the interface on both routers.

```
R1(config-if)# no shutdown
R2(config-if)# no shutdown
```

The link came up immediately, confirmed by a successful ping from R1 to R2 (Figure 4).

```
R1#ping 172.16.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.0.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 22/44/53 ms
```

Figure 4: R1 to R2 link restored

2.2 DHCP not assigning addresses

The DHCP pools existed on R1 but clients were not leasing addresses correctly. Reviewing the pool status confirmed the scopes and exclusions (Figure 5).

```
R1#show ip dhcp pool

Pool R1_LAN10 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Excluded addresses             : 2
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 172.16.10.1          172.16.10.1      - 172.16.10.254     0    / 2     / 254

Pool R1_LAN11 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Excluded addresses             : 2
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 172.16.11.1          172.16.11.1      - 172.16.11.254     0    / 2     / 254
```

Figure 5: R1 DHCP pool status

Fix: corrected the pool definitions with the right network, gateway, and exclusions for both subnets.

```
R1(config)# ip dhcp excluded-address 172.16.10.1 172.16.10.10
R1(config)# ip dhcp excluded-address 172.16.11.1 172.16.11.10
R1(config)# ip dhcp pool R1_LAN10
R1(config-dhcp)# network 172.16.10.0 255.255.255.0
R1(config-dhcp)# default-router 172.16.10.1
R1(config)# ip dhcp pool R1_LAN11
R1(config-dhcp)# network 172.16.11.0 255.255.255.0
R1(config-dhcp)# default-router 172.16.11.1
```

Both PCs then leased an address successfully.



Figure 6: PC1 — DHCP request successful (same for PC2)

2.3 No route to the internet

Even with the WAN link up, R1 had no route beyond its directly connected networks — `show ip route` showed no default route (Figure 7).

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B -
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C       172.16.0.0/30 is directly connected, Serial0/0/0
C       172.16.10.0/24 is directly connected, FastEthernet0/0
C       172.16.11.0/24 is directly connected, FastEthernet0/1
R       172.16.20.0/24 [120/1] via 172.16.0.2, 00:00:05, Serial0/0/0
```

Figure 7: R1 routing table — no default route

Fix: added a default route pointing to R2.

```
R1(config)# ip route 0.0.0.0 0.0.0.0 172.16.0.2
```

R1 could then reach the ISP (Figure 8), and R2's own link to the ISP was already confirmed working (Figure 8).

```
R1#ping 209.165.201.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.201.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 19/85/113 ms
```

Figure 8: R1 is reaching the ISP like R2

2.4 Dynamic NAT only worked for one subnet

R2's NAT access-list only permitted the 172.16.10.0/24 subnet, so PC2 (on 172.16.11.0/24) could not get translated (Figure 9).

```
R2#show access-lists
Standard IP access list NAT_ACL
    10 permit 172.16.10.0 0.0.0.255
```

Figure 9: NAT access-list — only one subnet permitted

Fix: rewrote the access-list to cover the whole internal range.

```
R2(config)# no access-list NAT_ACL
R2(config)# ip access-list standard NAT_ACL
R2(config-std-nacl)# permit 172.16.0.0 0.0.255.255
R2(config)# ip nat inside source list NAT_ACL interface Serial0/0/1
    overload
```

Dynamic mappings began appearing in the NAT statistics (Figure 10), and a PC could now ping the ISP (Figure 11).

```
R2(config)#ip access-list standard NAT_ACL
R2(config-std-nacl)#permit 172.16.0.0 0.0.255.255
R2(config-std-nacl)#exit
R2(config)#show ip nat statistics
           ^
% Invalid input detected at '^' marker.

R2(config)#exit
R2#show ip nat statistics
Total translations: 1 (1 static, 0 dynamic, 0 extended)
Outside Interfaces: Serial0/0/1
Inside Interfaces: FastEthernet0/0 , Serial0/0/0
Hits: 0  Misses: 15
Expired translations: 0
Dynamic mappings:
R2#show access-lists
Standard IP access list NAT_ACL
    10 permit 172.16.0.0 0.0.255.255
```

Figure 10: Corrected access-list and active NAT translations

```
C:\>ping 209.165.201.2

Pinging 209.165.201.2 with 32 bytes of data:

Reply from 209.165.201.2: bytes=32 time=117ms TTL=253
Reply from 209.165.201.2: bytes=32 time=2ms TTL=253
Reply from 209.165.201.2: bytes=32 time=2ms TTL=253
Reply from 209.165.201.2: bytes=32 time=2ms TTL=253

Ping statistics for 209.165.201.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 117ms, Average = 30ms
```

Figure 11: Client reaching the ISP through NAT

2.5 Web server unreachable from outside

The DMZ web server (172.16.20.254) had no static NAT entry, so it was not reachable from the public side.

Fix: added a static translation to its public address.

```
R2(config)# ip nat inside source static 172.16.20.254 209.165.201.30
```

The translation table confirmed the static entry (Figure 12).

```
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip nat inside source static 172.16.20.254 209.165.201.30
R2(config)#show ip nat translations
           ^
% Invalid input detected at '^' marker.
           |
R2(config)#exit
R2#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
---  209.165.201.30    172.16.20.254      ---                ---
```

Figure 12: Static NAT entry active

The ISP still needed a route to the server's public address, since it only knew the /30 transit subnet:

```
ISP(config)# ip route 209.165.201.30 255.255.255.255 209.165.201.1
```

With that in place, the ISP successfully pinged the web server's public IP, confirming inbound access works end to end.

```
ISP#ping 209.165.201.30

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.201.30, timeout is 2 seconds:

%SYS-5-CONFIG_I: Configured from console by console
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 41/433/1016 ms
```

Figure 13: ISP reaching the web server through static NAT

3 Summary

Issue	**Fix**
Serial link down between R1 and R2	`no shutdown` on both ends
DHCP pools misconfigured	Corrected network, gateway, and exclusions
No default route on R1	Added `ip route 0.0.0.0 0.0.0.0 172.16.0.2`
Dynamic NAT ACL too restrictive	Widened ACL to 172.16.0.0/16
No inbound path to the web server	Added static NAT entry
ISP had no route to the server's public IP	Added host route on ISP

4 Conclusion

All six faults traced back to simple oversights: a disabled interface, an incomplete DHCP scope, a missing default route, an overly narrow ACL, and two missing NAT/routing entries.
Working through the OSI layers in order — physical, then routing, then NAT — made each fault easy to isolate before moving to the next.
At the end, PC1 and PC2 obtain addresses via DHCP, both reach the ISP through dynamic NAT, and the ISP reaches the DMZ web server through static NAT.